

- 6 (a) (i) amplitude = 7.6 mm allow 7.5 mm A1 [1]
- (ii) $180^\circ / \pi$ rad A1 [1]
- (iii) $v = f \times \lambda$
 $= 15 \times 0.8$ C1
 $= 12 \text{ m s}^{-1}$ A1 [2]
- (b) correct sketch with peak moved to the right B1
 curve moved by the correct phase angle / time period of 0.25 T B1 [2]
- (c) (i) zero (rad) A1 [1]
- (ii) antinode maximum amplitude,
 node zero amplitude / displacement A1 [1]

Page 4	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE AS/A LEVEL – May/June 2011	9702	23

(iii) 3 A1 [1]

(iv) horizontal line through central section of wave B1 [1]

7 (a) density in solids and liquids similar M1